Gaussian Splatting for Physical Inverse Problems

MS Thesis I — Progress Report for the Meeting of 10 November 2026

Muhammad Hussain Qureshi [Roll no.]
Advisor: [Advisor name]
Faculty of Sciences, Information Technology University, Lahore

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Abstract

We use Gaussian Splatting as a representation for physical fields that are observed only through a known forward model, using datasets and operators from InverseBench. 3D Gaussians reconstruct cells from scattered electromagnetic fields (inverse scattering), and 4D Gaussians with an MLP or Mamba deformation network reconstruct 2D Navier–Stokes flow over time from sparse observations. All three planned experiments run end to end, the code is in a structured GitHub repository, and the thesis document is drafted. This report summarises the method, the results (including negative ones), the new true 3D scattering model, and this week's main result: with a *bounded* density, Gaussians beat a regularised voxel grid under the true 3D model on every volume with about nine times fewer parameters. The questions for today's meeting are highlighted in orange boxes.

1 Status at a glance

Item	Status	Note
3D Gaussians on inverse scattering (2D and 3D)	done	2D cells, synthetic and real (BBBC024) 3D nuclei
4D Gaussians on Navier–Stokes	done	held-out frames, 100% and 10% observations
Mamba as the motion model	done	MLP vs. Mamba vs. bidirectional Mamba
True 3D scattering model	done	sensors on a sphere (Section 4)
Bounded Gaussian density	done	Gaussians beat the voxel grid in true 3D (Section 5)
Sharper (generalised) Gaussian kernels	done	did not help under the physics loss
More samples with spread (10 cells)	done	paired comparison, mean ± std
GitHub repository (standard structure)	done	to be transferred to the thesis account
Thesis document (LaTeX, 41 pages)	draft	all required chapters, all results
Slides and Open House poster (A0)	draft	updated with all results

Main results to discuss

1. **True 3D scattering** (your request) is implemented and verified: 3D Born model, 60 transmitters and up to 360 receivers on a sphere.
2. **Bounded Gaussians win in true 3D:** rendering $1 - e^{-f}$ instead of f makes Gaussians beat a voxel grid with the same regularisation on every volume (real BBBC024 nuclei: up to +1.1 dB, synthetic: up to +4.3 dB) with ≈60k instead of 524k parameters.
3. **4D:** Mamba with temporal smoothness beats the 4DGS-style MLP by 2.4 dB on unseen frames when only 10% of pixels are observed.
4. **Honest limits:** with better-conditioned measurements (slice-wise 3D, 2D with many receivers) the regularised grid is still slightly better; unbounded or sharper Gaussians do not help.

Questions for today (details in Section 7)

1. **Is this true 3D model what you wanted, and is it sufficient for Thesis I?**
2. **Is the 4D task acceptable as is, or should we add physics (Navier–Stokes residual / initial condition)?**
3. **What should the next two weeks focus on?**
4. **What is the exact list of Thesis I targets, and is there a required template?**
5. **Thesis title and framing: confirm, and how (if at all) to connect it to 4D LangSplat.**

2 Method

Representation. A field f on a grid is a sum of anisotropic Gaussians,

$$f_\Theta(\boldsymbol{x}) = \sum_{i=1}^{N} a_i \exp\Big(-\tfrac{1}{2}\big[(\boldsymbol{x}-\boldsymbol{\mu}_i)^\top \Sigma_i^{-1} (\boldsymbol{x}-\boldsymbol{\mu}_i)\big]^\beta \Big), \qquad \Sigma_i = R_i S_i^2 R_i^\top, \tag{1}$$

with position $\boldsymbol{\mu}_i$, scales S_i, rotation R_i (angle in 2D, quaternion in 3D) and amplitude $a_i \geq 0$ (signed for vorticity). $\beta = 1$ is the usual Gaussian; $\beta > 1$ gives flatter tops and sharper edges. Each Gaussian is evaluated only in a local window around its centre, as in 3DGS.

Physics instead of a camera. The camera rasteriser of 3DGS is replaced by the forward operator $\mathcal{A}$ of the problem (Figure 1). Parameters are fitted with Adam to

$$\min_\Theta \ \frac{\|\mathcal{A} f_\Theta - \boldsymbol{y}\|_2^2}{\|\boldsymbol{y}\|_2^2} + \lambda \overline{f_\Theta}, \tag{2}$$

where the second term is a sparsity penalty on the mean intensity. New Gaussians are added where the back-projected residual $\mathrm{Re}\,\mathcal{A}^H(\boldsymbol{y} - \mathcal{A} f_\Theta)$ is largest; Gaussians with negligible amplitude are pruned. The baseline optimises every pixel or voxel directly with the same loss.

Inverse scattering. The linear (Born) model of InverseBench: $y_{t,r} = \Delta x\, \Delta y \sum_p f_p\, \overline{u^{\mathrm{inc}}_{p,t}}\, G_{p,r}$, with 20 transmitters and 360, 180 or 60 receivers on a circle (our implementation matches InverseBench to 3×10^{-7}). For 3D volumes the operator was first applied slice by slice. The new *true 3D* model (Section 4) uses 3D Green's functions with sensors on a sphere.

4D Gaussians for flow. Canonical Gaussians are moved over time by a deformation network $\Delta\boldsymbol{\mu}_i(t), \Delta s_i(t), \Delta a_i(t) = D_\phi(\boldsymbol{\mu}_i, t)$: an MLP with Fourier features (as in 4D Gaussian Splatting) or a Mamba selective state-space model that reads each Gaussian's trajectory over all time steps as a sequence (causal or bidirectional). A temporal smoothness term $\lambda_T \sum_t \|D_\phi(\cdot, t{+}1) - D_\phi(\cdot, t)\|^2$ links observed and unobserved times.

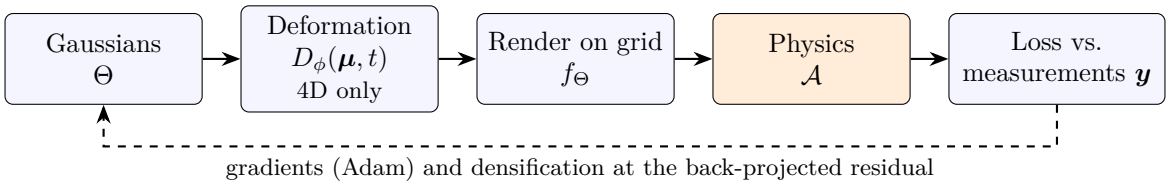


Figure 1: Pipeline: the camera of Gaussian Splatting is replaced by the physics operator.

3 Results so far

All runs: NVIDIA T4, 3000 iterations, InverseBench noise level. PSNR in dB / SSIM, means over samples.

3.1 2D inverse scattering (InverseBench cells)

Without regularisation, adding Gaussians made results *worse*: they grew ghost copies of the cell in regions the receivers cannot see (the operator's null space, Figure 2). A sparsity term removes the ghosts. With it, Gaussians beat the plain pixel grid in SSIM at 60 receivers, but a pixel grid given the same sparsity term matches or beats the Gaussians (Table 1): the gain comes from the prior, not from the representation.

Table 1: 2D inverse scattering, 5 cells (PSNR / SSIM).

Method	360 receivers	180 receivers	60 receivers
Gaussians, no sparsity (3 cells)	30.7 / 0.959	30.4 / 0.957	26.5 / 0.814
Gaussians + sparsity ($\lambda = 0.3$)	30.8 / 0.980	30.6 / 0.979	27.1 / 0.924
Pixel grid	**36.1** / 0.984	**34.7** / 0.974	**28.6** / 0.886
Pixel grid + sparsity ($\lambda = 0.3$)	33.2 / **0.985**	32.5 / **0.983**	27.7 / **0.933**

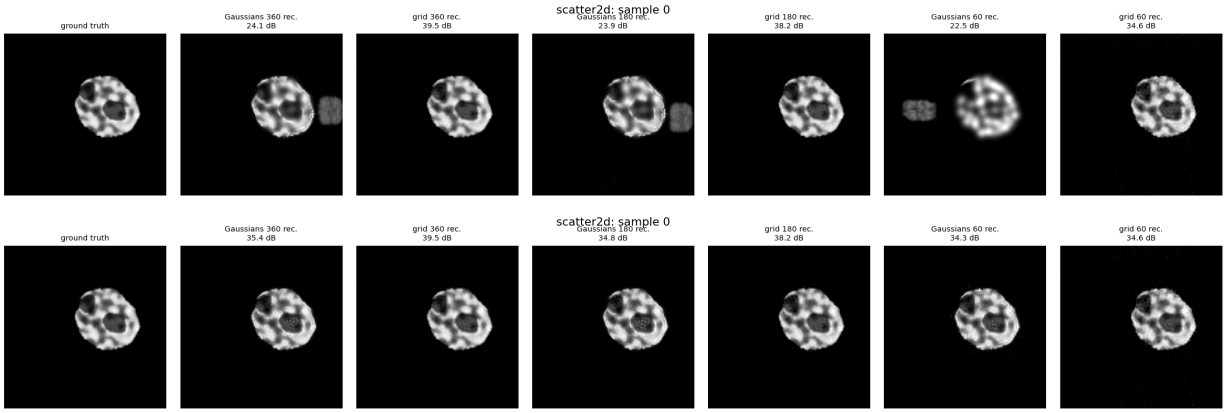


Figure 2: Cell 0. Top: aggressive densification, no sparsity: a ghost copy appears beside the cell. Bottom: with sparsity the ghost is gone. Columns: truth, then Gaussians and pixel grid at 360, 180 and 60 receivers.

3.2 3D inverse scattering (slice-wise operator)

On smooth synthetic nuclei, Gaussians barely change as receivers drop (40.8 → 40.9 dB) while the voxel grid falls to 37.3 dB and becomes speckled. On real BBBC024 nuclei, which are binary masks with sharp boundaries, Gaussians stay near 27–28 dB while voxels reach 35–42 dB: smooth kernels soften the edges (Table 2, Figure 3).

Table 2: 3D inverse scattering, $32 \times 128 \times 128$ volumes, 3 volumes (PSNR / SSIM).

Volumes	Method	360 receivers	180 receivers	60 receivers
Synthetic	Gaussians	40.8 / **0.994**	40.6 / **0.994**	**40.9 / 0.988**
	Voxel grid	**44.1** / 0.987	**41.5** / 0.977	37.3 / 0.932
BBBC024	Gaussians	27.9 / 0.967	27.6 / 0.965	27.0 / 0.943
	Voxel grid	**41.8** / 0.989	**40.3** / 0.980	**35.3** / 0.936
	Gaussians + sparsity	28.4 / 0.974	28.4 / 0.974	27.4 / 0.955
	Voxel grid + sparsity	38.5 / **0.996**	38.0 / **0.995**	34.4 / **0.971**

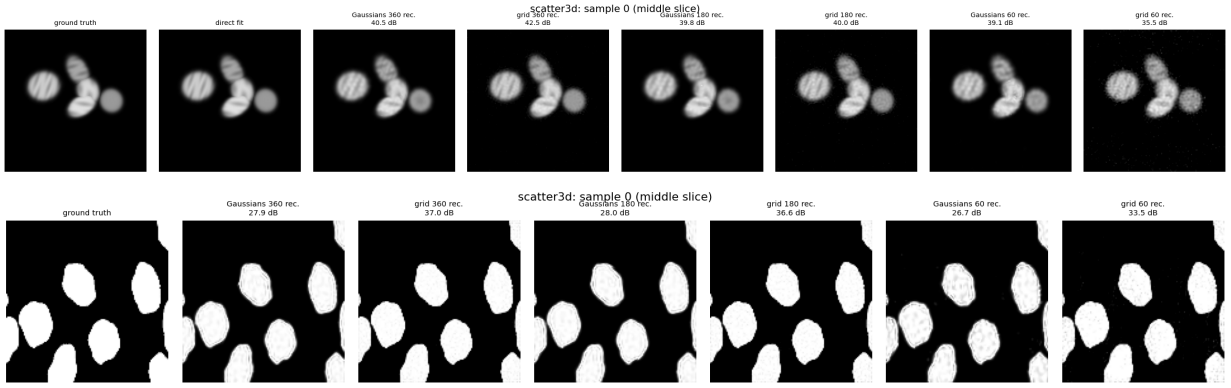


Figure 3: Middle slices. Top: synthetic volume (truth, direct fit, then Gaussians and voxels at 360/180/60 receivers); voxels become speckled at 60 receivers. Bottom: BBBC024 volume; Gaussians soften the sharp boundaries.

3.3 4D Navier–Stokes

Static Gaussians reach only 17 dB; a deformation network adds about 10 dB. Without regularisation, Mamba fits observed frames much better than the MLP but does not generalise better to unseen times. With temporal smoothness, Mamba beats the MLP by 2.4 dB on unseen frames when only 10% of pixels are observed, on each of the three sequences (Table 3, Figure 4). The bidirectional variant gives no further gain and is twice as slow.

Table 3: 4D Navier–Stokes, PSNR (dB) on held-out frames; observed-frame PSNR in brackets; 3 sequences.

Observed pixels	Static	MLP	Mamba	Bidirectional Mamba
100%	17.0 (16.8)	27.6 (30.3)	**27.9** (36.1)	–
100% + smoothness	–	27.4 (30.6)	27.2 (35.2)	26.4 (36.5)
10%	16.0 (15.9)	23.2 (24.0)	23.1 (27.7)	–
10% + smoothness	–	23.0 (24.1)	**25.4** (28.9)	25.1 (29.6)
Time per sequence	3 min	3 min	15–28 min	52 min

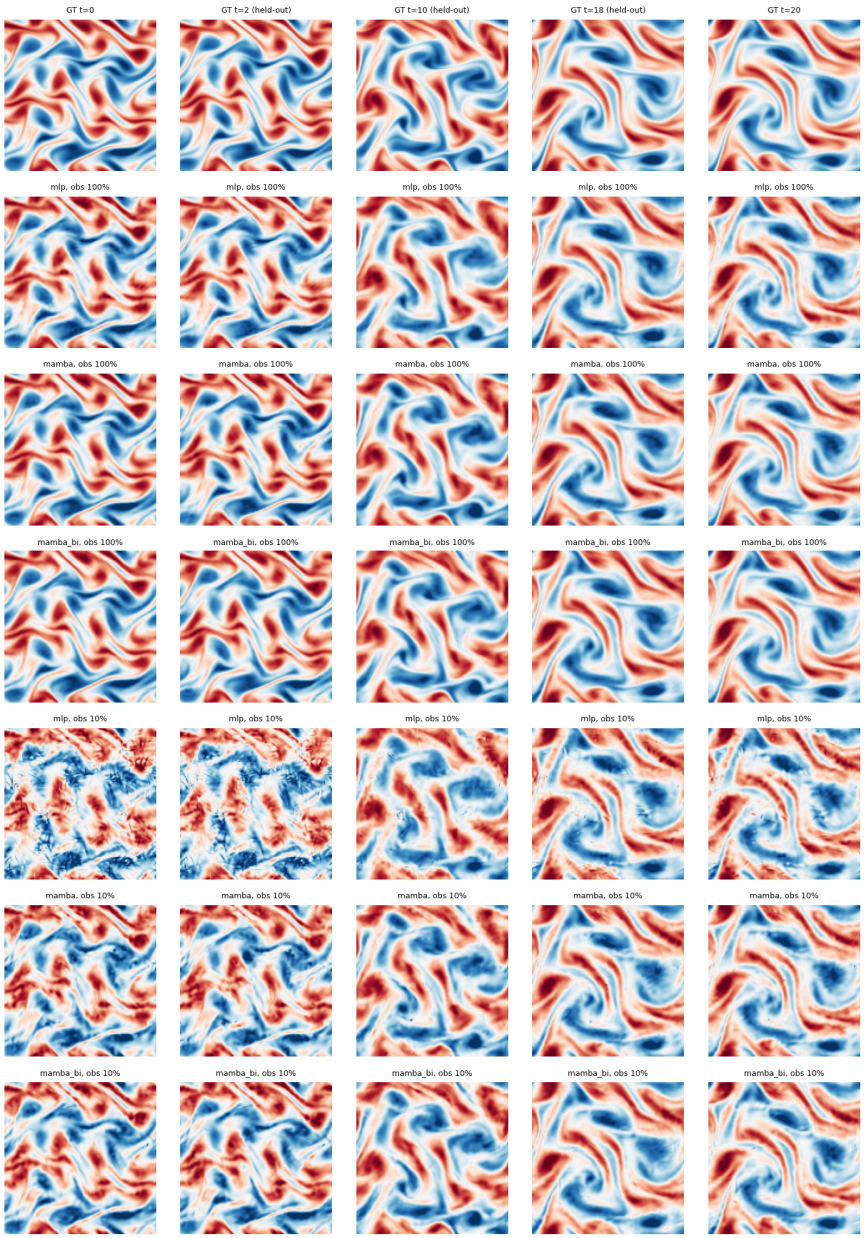


Figure 4: Vorticity of sequence 0 at $t = 0, 2^*, 10^*, 18^*, 20$ (* never observed). Rows: truth; MLP, Mamba and bidirectional Mamba with 100% and then 10% of pixels observed. With 10% the MLP shows streaks; Mamba stays coherent.

4 True 3D model, sharper kernels, ten cells

True 3D scattering. Following your request, the slice-wise operator is replaced by a first-order Born model in 3D,

$$y_{t,r} = k^2 \, \Delta V \sum_p f_p \, G(\|\boldsymbol{x}_t - \boldsymbol{x}_p\|) \, G(\|\boldsymbol{x}_r - \boldsymbol{x}_p\|), \qquad G(d) = \frac{e^{\mathrm{i}kd}}{4\pi d}, \tag{3}$$

with the same voxel size, wavelength and sensor radius as InverseBench, 60 transmitters and 360/180/60 receivers spread evenly over a sphere around the volume, and noise at the SNR of the 2D benchmark. The implementation matches the analytic response of a point scatterer to 10^{-7} and passes an adjoint test. With about ten times fewer measurements than the slice-wise model, the voxel grid is better than *unbounded* Gaussians in every setting (Table 4). On BBBC024, Gaussians

leave a bright rim at each boundary and under-fill the interior (Figure 5); the voxel grid benefits from being clipped to $[0, 1]$, a strong prior for binary objects that Gaussians lack. Gaussians are, however, 11–12 times more compact.

Table 4: True 3D scattering, $32 \times 128 \times 128$, sparsity 0.3 for both methods, 3 volumes (PSNR / SSIM).

Volumes	Method	Params	360 rec.	180 rec.	60 rec.
Synthetic	Gaussians	42k	37.6 / 0.977	36.6 / 0.955	32.1 / 0.852
	Voxel grid	524k	**42.1 / 0.985**	**39.9 / 0.976**	**38.2 / 0.966**
BBBC024	Gaussians	47k	25.6 / 0.960	24.9 / 0.940	20.9 / 0.772
	Gaussians, $\beta = 4$	51k	24.2 / 0.954	22.6 / 0.927	16.3 / 0.722
	Voxel grid	524k	**32.5 / 0.983**	**30.5 / 0.956**	**25.6 / 0.875**

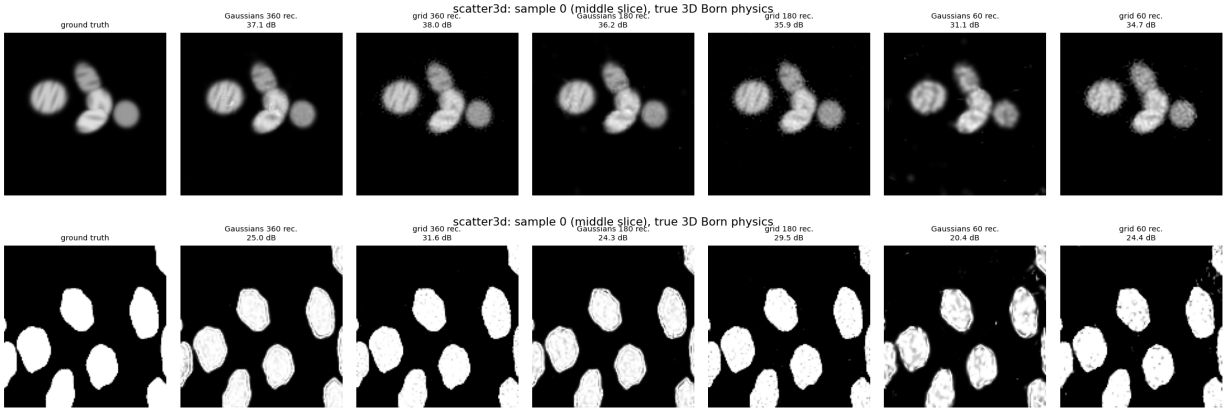


Figure 5: True 3D model, middle slices. Top: synthetic nuclei. Bottom: BBBC024. Columns: truth, then Gaussians and voxel grid at 360, 180 and 60 receivers.

Sharper kernels. Generalised Gaussians ($\beta = 4$) fit binary masks about 3 dB better when fitted directly, but under the physics loss they are slightly worse (slice-wise BBBC024: 28.1 / 28.0 / 25.5 dB vs. 28.4 / 28.4 / 27.4 dB).

Ten cells, paired. In 2D, the pixel grid with the same sparsity term beats Gaussians on all ten cells at every receiver count, by 0.6 ± 0.3, 2.3 ± 0.7 and 2.9 ± 0.8 dB at 60, 180 and 360 receivers. Against the plain grid, Gaussians with sparsity keep higher SSIM at 60 receivers (0.951 vs. 0.910).

5 This week: bounded Gaussians close the gap

The previous results pointed to one asymmetry: the voxel grid is clipped to $[0, 1]$ after every step, while the sum of Gaussians is unbounded and overshoots where Gaussians overlap (the bright rims in Figure 5). We render the bounded density

$$\tilde{f}_\Theta(\boldsymbol{x}) = 1 - \exp(-f_\Theta(\boldsymbol{x})) \in [0, 1), \tag{4}$$

the same saturation as opacity accumulation in 3DGS, and optionally initialise the Gaussians from the grid solution (B+G). Nothing else changes; the grid uses the same sparsity term.

Table 5: Bounded Gaussians (B), bounded and grid-initialised (B+G) vs. the grid with the same sparsity term. PSNR dB / SSIM.

Setting	Method	Params	360 rec.	180 rec.	60 rec.
True 3D, BBBC024 (3)	Gaussians (unbounded)	47k	25.6 / 0.960	24.9 / 0.940	20.9 / 0.772
	Gaussians B	47k	32.5 / **0.991**	31.3 / **0.983**	**26.8** / 0.903
	Gaussians B+G	60k	**32.8** / 0.990	**31.4** / 0.980	26.7 / **0.909**
	Voxel grid	524k	32.5 / 0.983	30.5 / 0.956	25.6 / 0.875
True 3D, synthetic (3)	Gaussians B+G	54k	**46.4 / 0.993**	**43.6 / 0.986**	**39.1** / 0.963
	Voxel grid	524k	42.1 / 0.985	39.9 / 0.976	38.2 / **0.966**
Slice-wise, BBBC024 (3)	Gaussians B+G	60k	35.8 / 0.995	35.7 / 0.995	34.1 / **0.983**
	Voxel grid	524k	**38.5 / 0.996**	**38.0** / 0.995	**34.4** / 0.971
2D cells (10)	Gaussians B	3.2k	33.7 / 0.988	33.3 / 0.988	30.4 / 0.957
	Gaussians B+G	4.0k	34.4 / 0.989	34.0 / **0.989**	**30.5** / 0.957
	Pixel grid	16.4k	**35.0 / 0.990**	**34.2** / 0.988	30.0 / 0.957

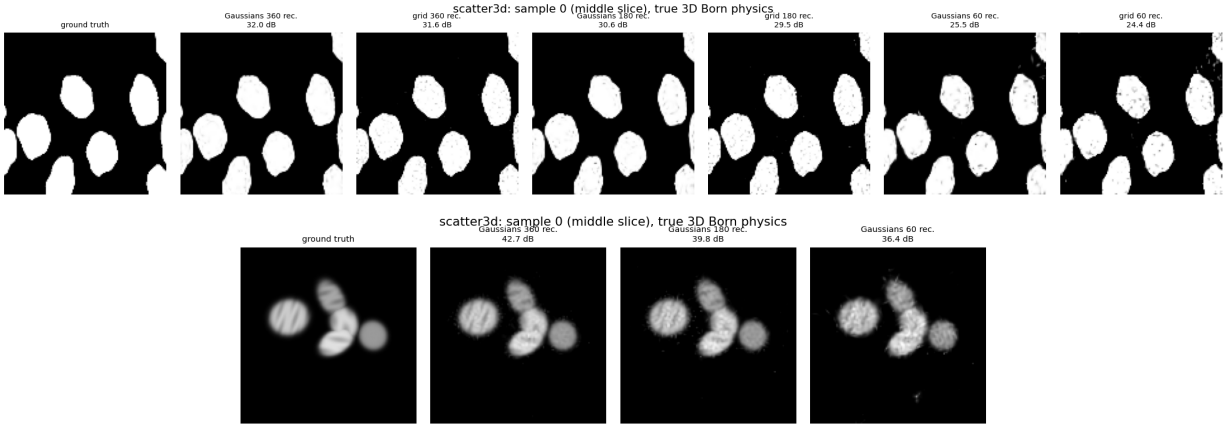


Figure 6: True 3D model, bounded grid-initialised Gaussians. Top: BBBC024 volume 0 (truth, then Gaussians and voxel grid at 360/180/60 receivers); the rims of Figure 5 are gone. Bottom: synthetic volume 0, truth and Gaussians (voxel grid: 38.0 / 35.9 / 34.7 dB).

Paired comparison. In true 3D, B+G Gaussians beat the voxel grid on every volume at every receiver count: $+0.26 \pm 0.07$, $+0.97 \pm 0.09$ and $+1.12 \pm 0.10$ dB on BBBC024 and +4.3, +3.7, +0.9 dB on synthetic nuclei (360/180/60 receivers). In 2D, bounded Gaussians win at 60 receivers on 6 of 10 cells (+0.4 dB, within the spread) and lose by 0.2–1.4 dB at 180 and 360.

Summary. Bounding the density is essential for Gaussians on quantities with a known range, and matters more than the kernel shape. Gaussians help most where the problem is most underdetermined (true 3D: about 21.6k measurements for 524k unknowns), with an order of magnitude fewer parameters; with plentiful measurements a regularised grid remains slightly better. In 4D, Mamba helps when observations are sparse.

6 Code and documents

- Repository: `https://github.com/muhammadhussainqureshi/thesis-gs` (`src/`, `bin/`, `configs/`, `notebooks/`, `results/`, `tests/`, `docs/`). Every run stores its exact settings, per-sample metrics and figures. To be transferred to the thesis account with the required collaborators.

- Thesis document: LaTeX in `docs/thesis/` (background, literature review, problem and scope, methodology, formulation, setup, results, discussion and Thesis II plan).

7 Questions for the meeting

Please advise on

1. **True 3D model.** Is the 3D Born model with sensors on a sphere (Section 4) what you had in mind? Is first-order (Born) scattering sufficient for Thesis I, or do you expect multiple scattering?

2. **Framing of the scattering result.** Main claim: *bounded Gaussians are a compact (about 9× fewer parameters) and more accurate representation in strongly under-determined 3D problems, and a regularised grid remains better with plentiful data.* Do you agree with this framing for the thesis and the Open House?

3. **4D task.** We reconstruct the flow from sparse observations of the vorticity itself. Is this acceptable, or should we add physics (Navier–Stokes residual as a loss) or recover the initial condition through the solver, as InverseBench does?

4. **Priorities for the remaining weeks.** Which first: (a) physics-informed 4D model, (b) more samples / full test sets with confidence intervals, (c) extending the literature review (currently 33 references), (d) the bounded density in the 4D model?

5. **Thesis I requirements.** The exact list of targets you expect; the required thesis template and length; the thesis title.

6. **Connection to 4D LangSplat.** Should the thesis mention the original 4D LangSplat direction (e.g. as Thesis II future work), or leave it out?

7. **Administration.** Please confirm the email address to add you as a collaborator on the thesis GitHub repository.